EduDOCS

An Automated Framework for Academic Document Compliance and Evaluation

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INTRODUCTION

- **EduDOC** is an academic document compliance and evaluation system designed to reduce manual effort in document verification while preserving teacher control.

- Academic submissions require **strict guideline adherence**

- Manual checking is **time-consuming and inconsistent**

- EduDOC automates **format, structure, and rule validation**

- Supports **Teacher, Student, and Admin** roles in a controlled workflow

ABSTRACT

- Academic document evaluation demands **strict compliance, originality, and submission discipline**
- Existing practices are **largely manual**, increasing faculty workload and inconsistency
- EduDOC automates **guideline verification, format checking, and rule enforcement**
- Teachers define guidelines while retaining **full evaluation authority**
- The system ensures **fairness, transparency, and auditability** in academic submissions

PROBLEM STATEMENT

Manual academic document evaluation is time-consuming, error-prone, and inconsistent, lacking an automated system to ensure strict guideline compliance while preserving faculty control.

PROPOSED SYSTEM

- A **web-based automated framework** for academic document compliance
- Teachers **define and control guidelines** for each academic activity
- Students submit documents within **rule-bound and time-bound workflows**
- System performs **format, structure, and policy validation**
- Ensures **transparent evaluation and audit-ready tracking**

LITERATURE REVIEW

Title & Year	Author(s) & Publication	Algorithm Used	Features Extracted / Methodology	Performance Metric	Results / Findings
ARCEAK: An Automated Rule Checking Framework Enhanced with Architectural Knowledge (2025)	Junyong Chen et al., arXiv Preprint	Rule-based checking + Transformer-assisted rule extraction	Rule extraction from structured documents, rule validation, compliance reasoning	Case-level evaluation accuracy	Demonstrated effective automated rule compliance checking for structured documents
Stateless and Rule-Based Verification for Compliance Checking Applications (2022)	M.R. Besharati et al., arXiv	Logic-based rule verification	Formal rule modeling, stateless compliance validation	Correctness of rule evaluation	Achieved reliable and consistent compliance verification without state dependency
Machine Learning and Rule-Based Embedding Techniques for Document Categorization (2024)	Springer Journal	Hybrid ML + rule-based approach	Text feature extraction, rule embeddings, document structure analysis	Accuracy, Precision	Improved classification accuracy using hybrid rule–ML approach

Title & Year	Author(s) & Publication	Algorithm Used	Features Extracted / Methodology	Performance Metric	Results / Findings
Smart Compliance Checking Framework for BIM Standards (2024)	X. Zhu, Academic Research	NLP + Rule-based validation	Natural language rule interpretation, compliance checking pipeline	Rule satisfaction rate	Reduced manual compliance effort; validated complex standards automatically
On the Complexities of Testing for Compliance with Human-Centric Standards (2025)	Springer Book Chapter	Compliance testing models	Rule ambiguity handling, human-in-the-loop compliance verification	Qualitative compliance validation	Highlighted need for semi-automated systems with human confirmation
A Survey on AI-Based Plagiarism Detection Techniques (2024)	IEEE Access (Survey)	NLP similarity + ML models	N-gram similarity, semantic similarity, citation analysis	Precision, Recall, F1-score	Identified hybrid NLP + rule-based methods as most effective

MODULES OF PROPOSED SYSTEM

Guideline Definition & Configuration Module

Enables teachers to define, manage, and confirm academic document guidelines for each activity.

Document Submission & Control Module

Handles student submissions with enforced deadlines, limited attempts, and submission locking.

Automated Compliance Verification Module

Automatically checks document structure, format, and rule adherence against defined guidelines.

Plagiarism & Originality Analysis Module

Evaluates document originality to detect similarity and potential academic misconduct.

Evaluation & Scoring Module

Supports rubric-based assessment with time penalties while retaining full teacher control.

Audit Trail & Monitoring Module

Maintains complete logs of submissions, evaluations, and system activities for transparency.

Comparison

Dimension	Existing Systems	Proposed System (EduDOC)
System Scope	Focus on **isolated functionalities** (formatting or plagiarism only)	End-to-end academic document compliance and evaluation framework
Guideline Representation	Static templates or informal manual interpretation	Explicit, machine-readable guideline definitions with teacher validation
Compliance Verification	Semi-automated or manual inspection	Automated, rule-based compliance verification engine
Document Structure Analysis	Limited structural parsing	Hierarchical document structure extraction and validation
Plagiarism Detection	External or standalone similarity tools	Tightly integrated plagiarism and originality analysis module
Evaluation Methodology	Manual grading with subjective judgment	Rubric-driven, rule-assisted evaluation with human override
Deadline and Policy Enforcement	Administrative monitoring	System-enforced submission policies with temporal constraints
Transparency and Traceability	Minimal logging and limited auditability	Comprehensive audit trails covering submissions, evaluations, and rule execution
Consistency of Assessment	Evaluator-dependent variability	Deterministic rule execution ensuring uniform assessment criteria
Feedback Mechanism	Informal or delayed feedback	Structured, system-generated compliance and evaluation reports
Scalability and Maintainability	Tool-specific and difficult to extend	Modular, extensible architecture supporting future enhancements
Academic Integrity Support	Reactive plagiarism checks	Proactive integrity enforcement integrated into submission workflow

PROJECT PLAN

Phase	Duration	Date Range	Key Activities	Deliverables
Phase 1	Requirement Analysis & Planning	01 Jan – 05 Jan 2026	Problem definition, feasibility study, scope finalization, requirement gathering, role definition	Software Requirement Specification (SRS)
Phase 2	System Design	06 Jan – 12 Jan 2026	Database design, UML diagrams (Use Case, Class, Activity, Sequence, State), architecture design	Complete design documentation
Phase 3	Environment Setup	13 Jan – 15 Jan 2026	Django project setup, MySQL configuration, authentication setup, Git repository	Development environment ready
Phase 4	Core Module Development	16 Jan – 31 Jan 2026	User authentication, classroom & assignment modules, guideline management, submission workflow	Functional Teacher & Student modules
Phase 5	Compliance & Evaluation Engine	01 Feb – 10 Feb 2026	Format checking, plagiarism detection, rubric-based evaluation, late penalty logic	Automated validation & evaluation engine
Phase 6	Admin & Audit Modules	11 Feb – 15 Feb 2026	Admin dashboard, user monitoring, audit logs, notifications	Admin & audit functionality
Phase 7	Testing & Optimization	16 Feb – 19 Feb 2026	Unit testing, integration testing, bug fixing, performance optimization	Tested and stable system
Phase 8	Documentation & Final Submission	20 Feb – 23 Feb 2026	Project report, PPT preparation, diagram finalization, viva preparation	Final report, PPT, deployed system

TOOLS & TECHNOLOGIES

Frontend: HTML, CSS, JavaScript, Bootstrap

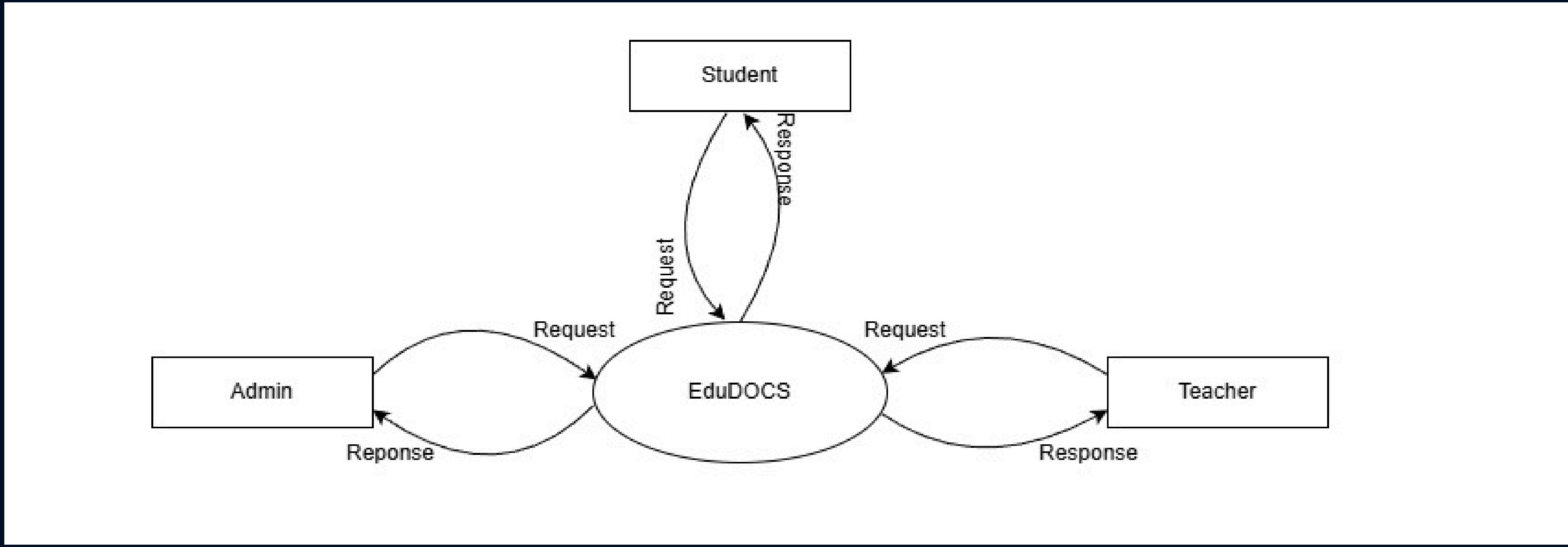
Backend: Python (Django Framework)

Database: MySQL

Document Processing: Python Libraries (python-docx, PDF parsing utilities)

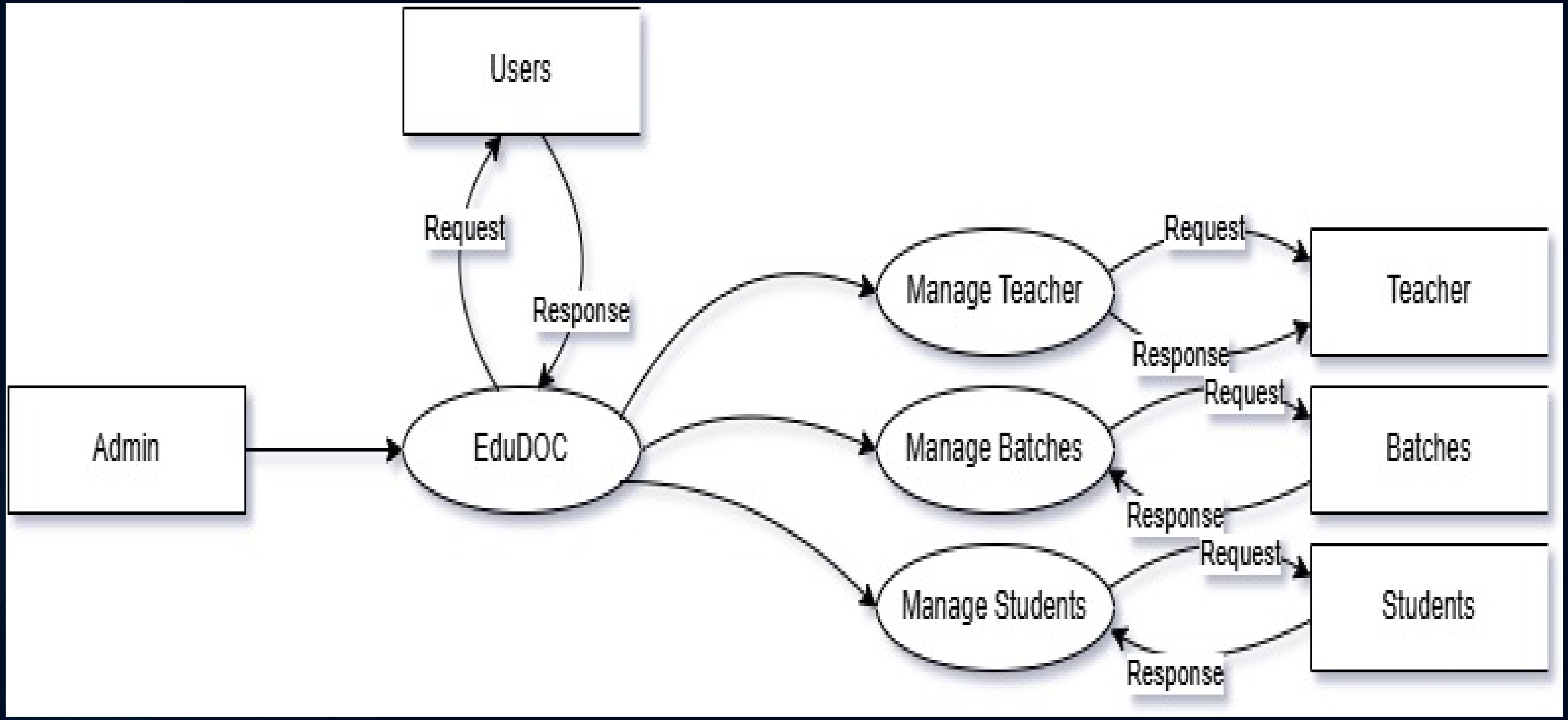
DIAGRAMS

DFD LEVEL 0

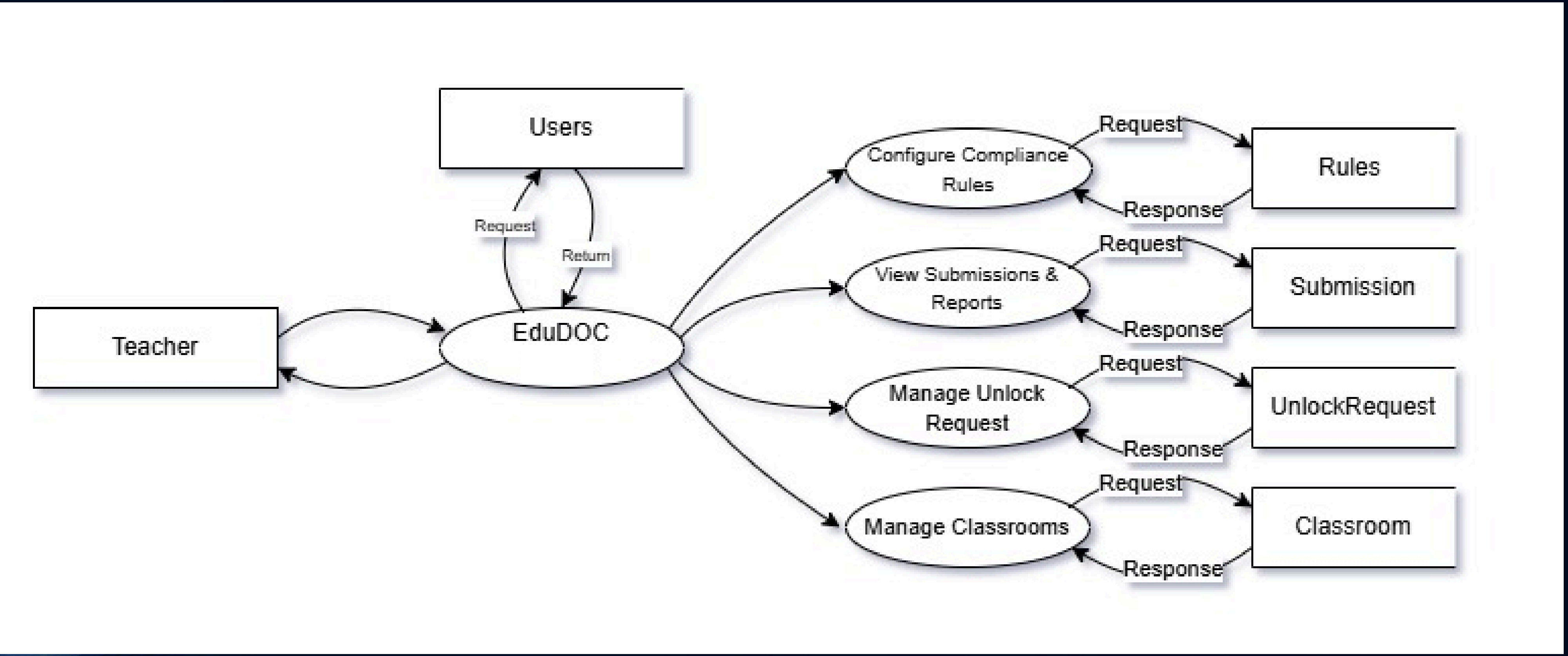


Student
Response
Request
Request
Request
Admin
EduDOCS
Teacher
Reponse
Response

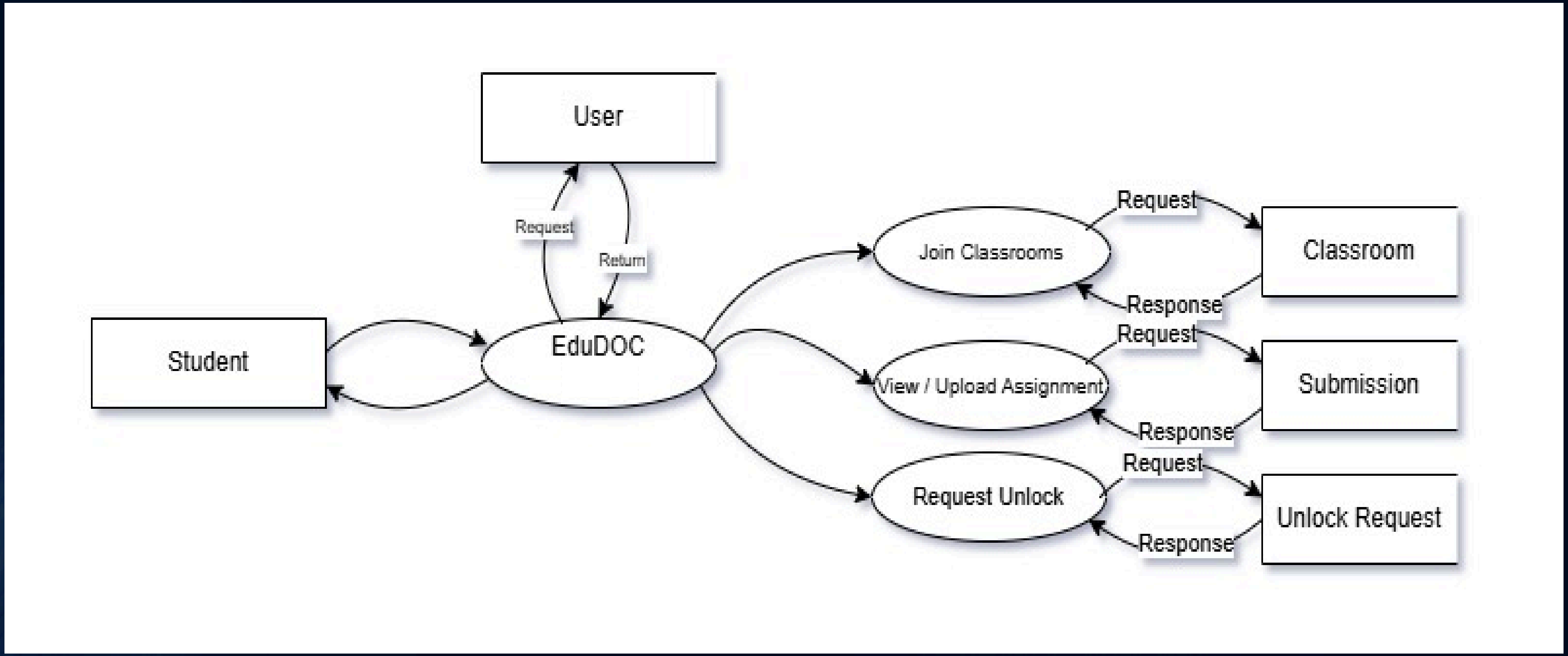
DFD LEVEL 1 ADMIN



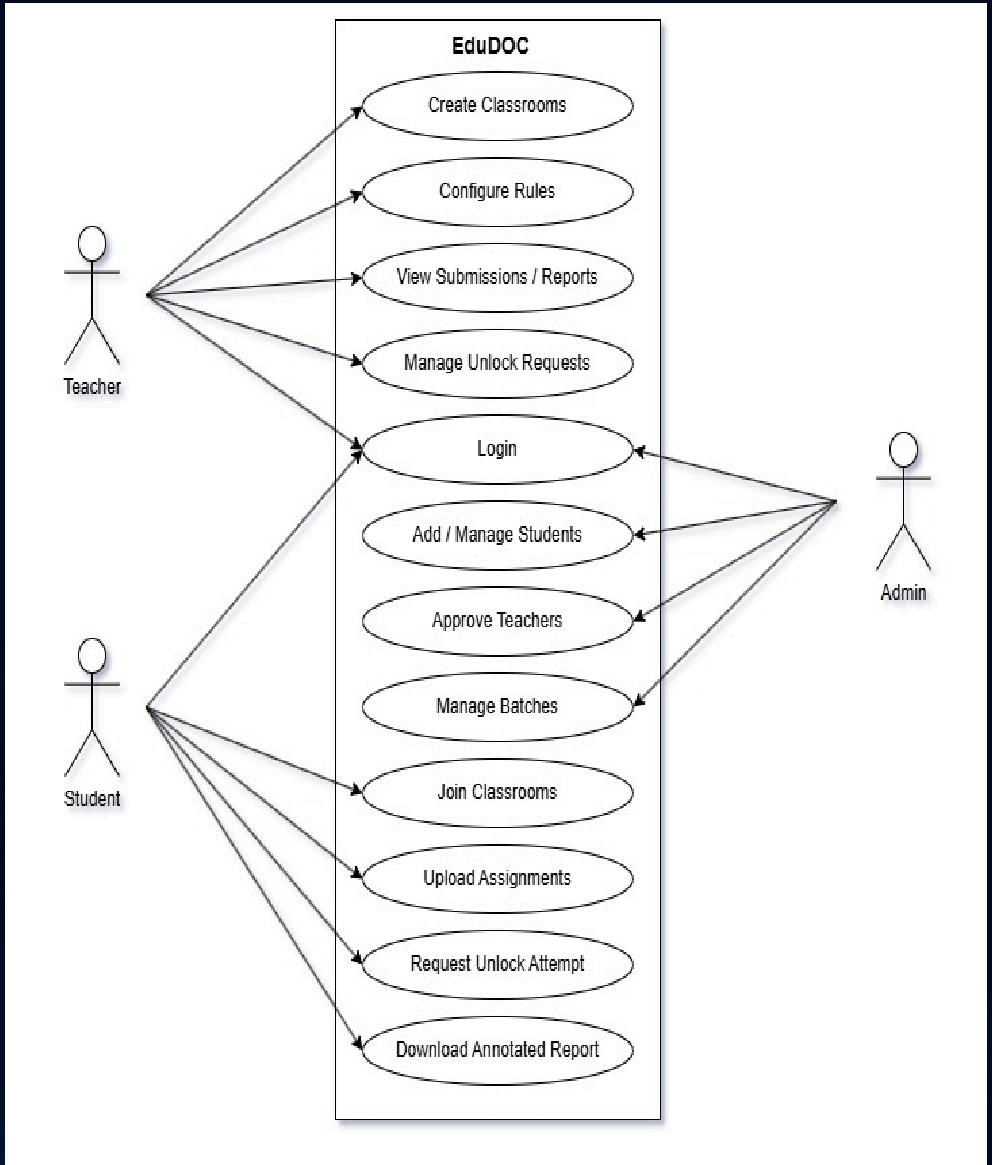
DFD LEVEL 1 TEACHER



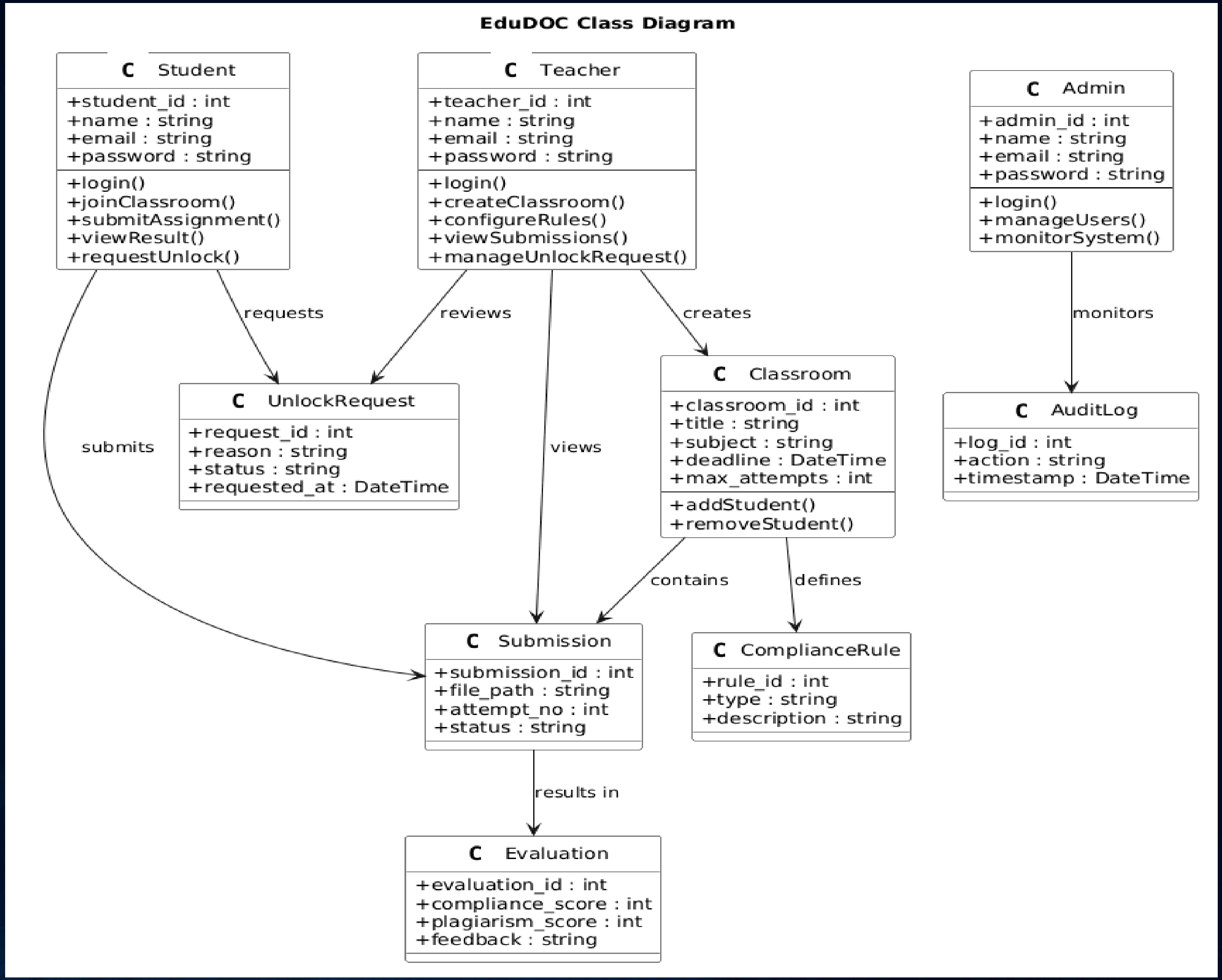
DFD LEVEL 1 STUDENT



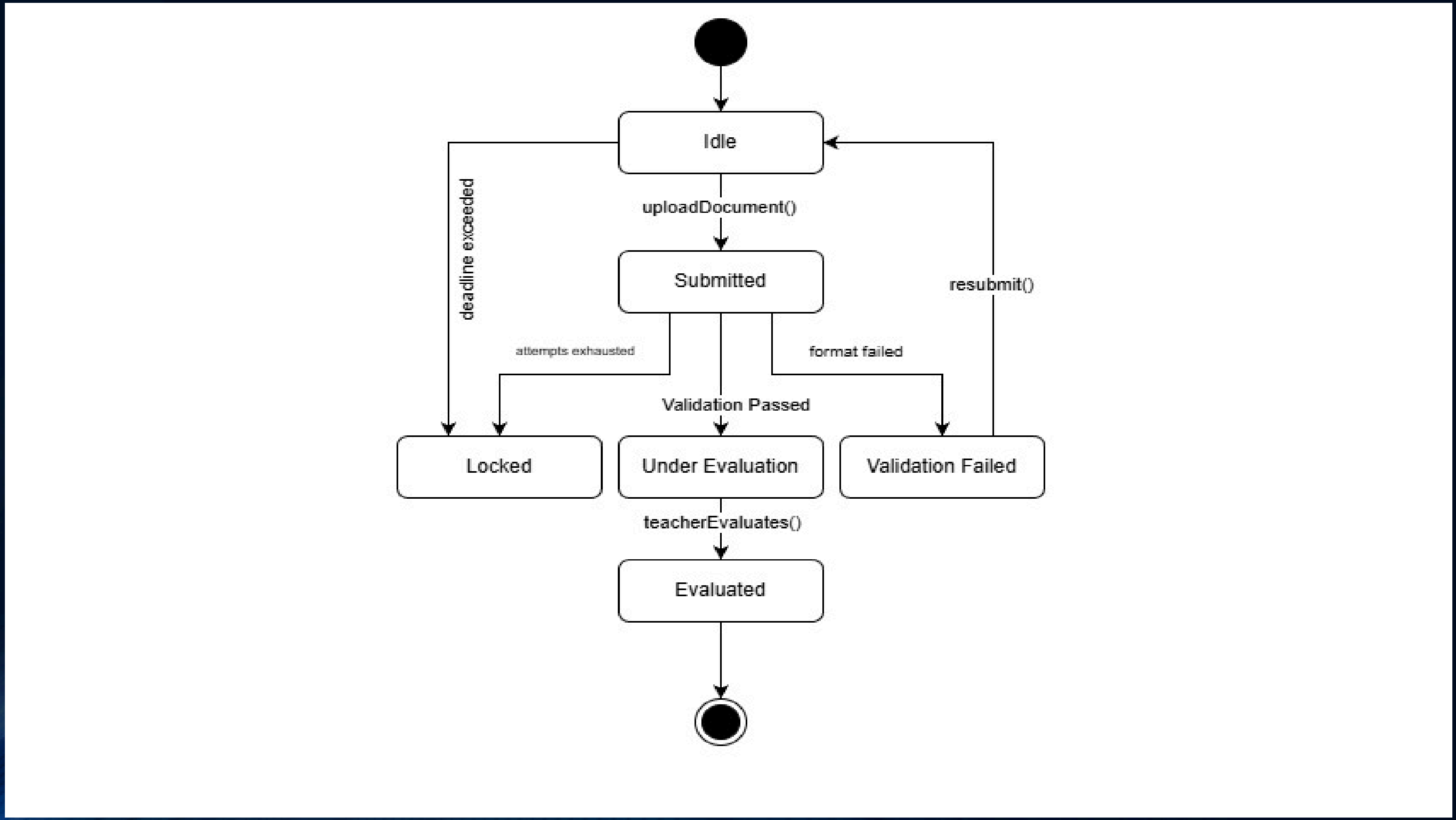
USE CASE DIAGRAM



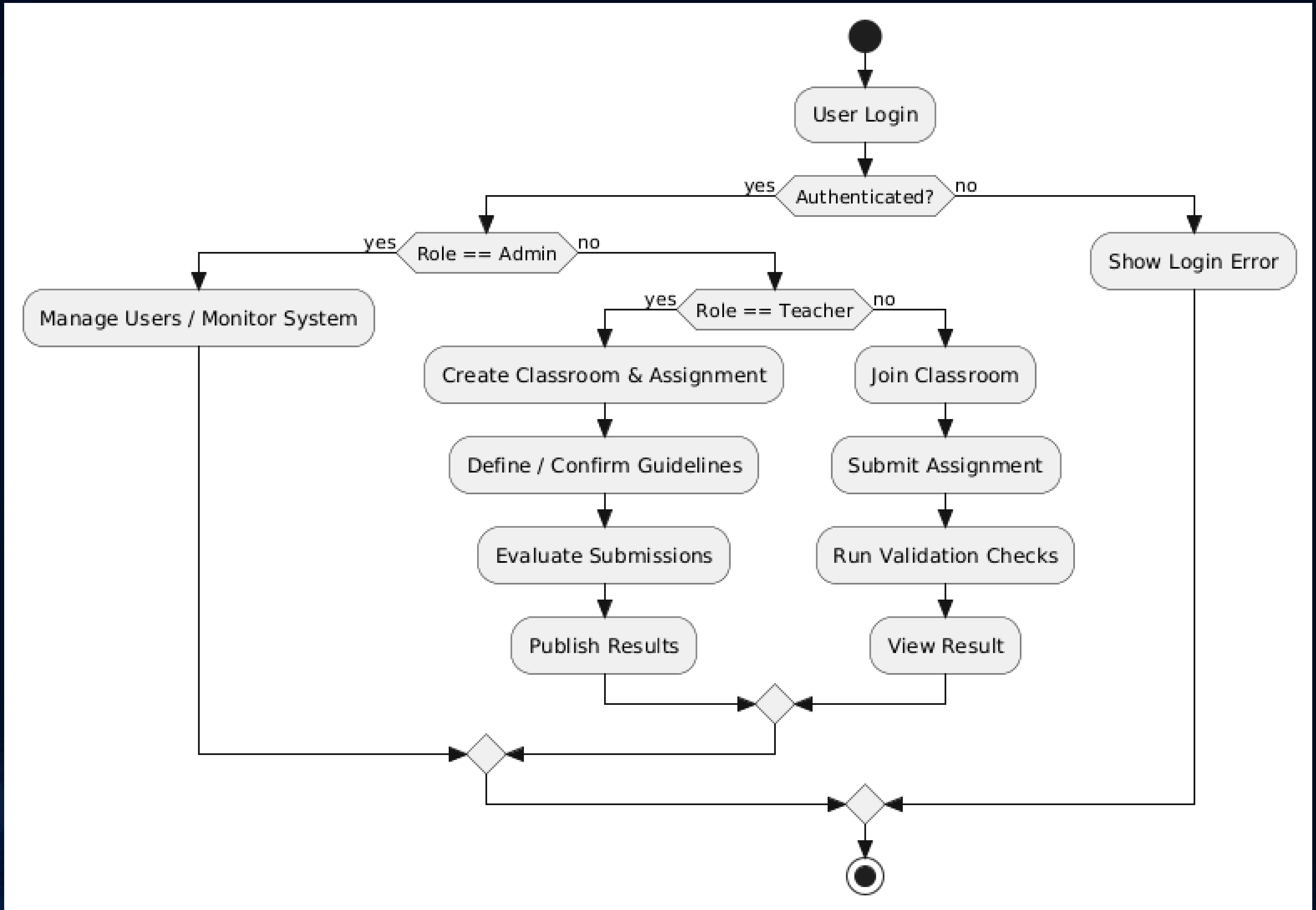
CLASS DIAGRAM



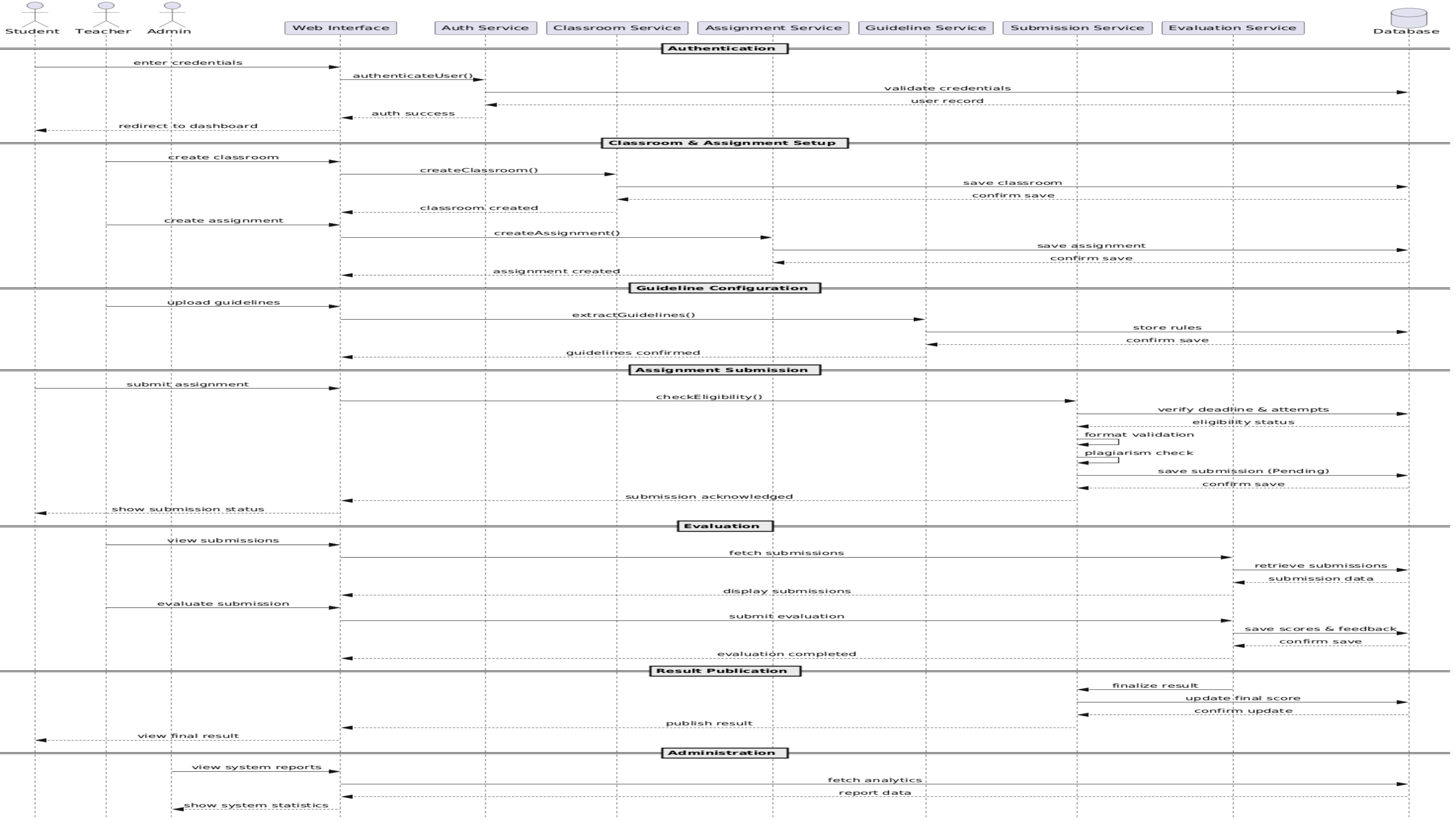
STATE DIAGRAM



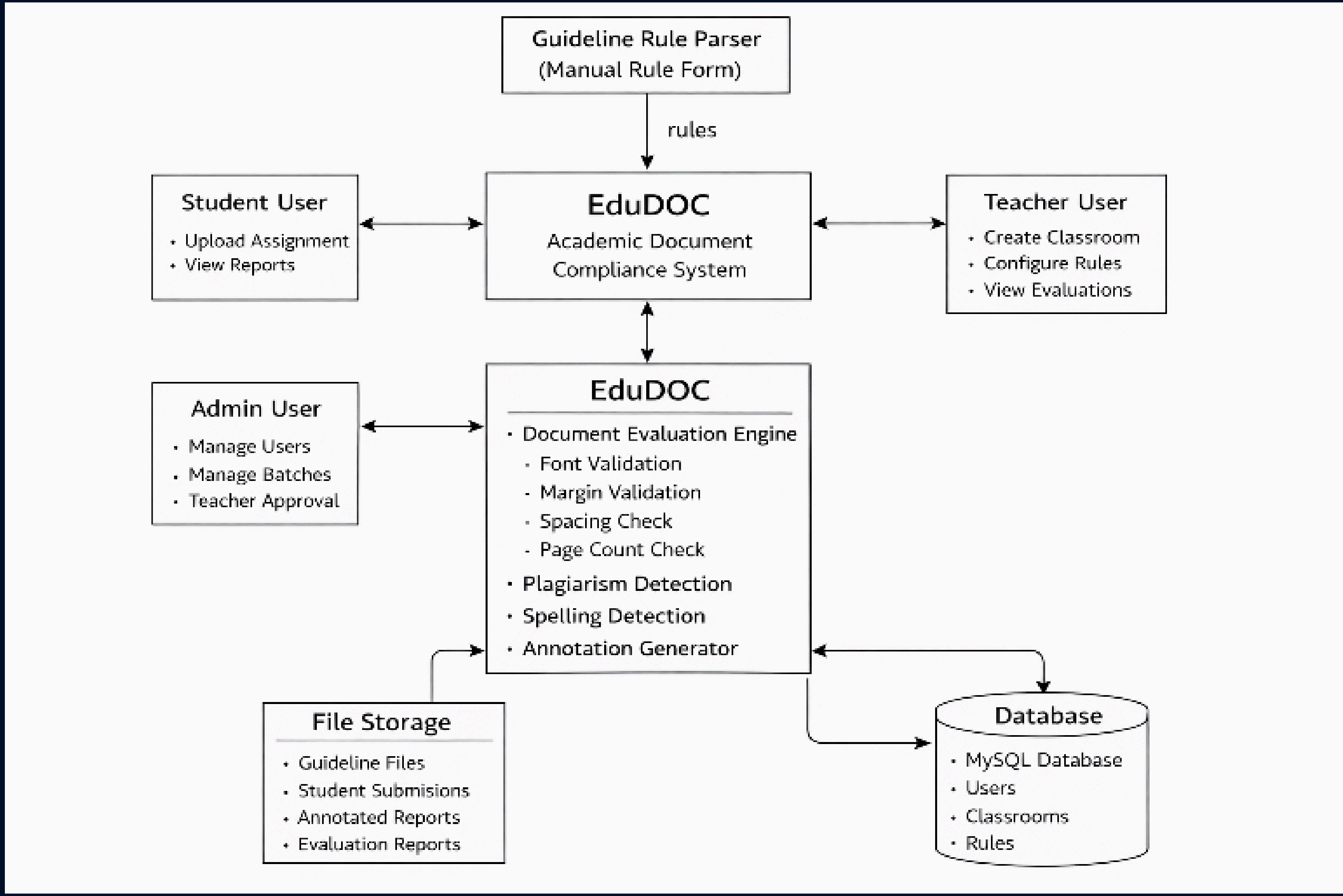
ACTIVITY DIAGRAM



SEQUENCE DIAGRAM


Student
Teacher
Admin
Web Interface
Auth Service
Classroom Service
Assignment Service
Guideline Service
Submission Service
Evaluation Service
Database
Authentication
enter credentials
authenticateUser()
validate credentials
user record
auth success
redirect to dashboard
Classroom & Assignment Setup
create classroom
createClassroom()
save classroom
confirm save
classroom created
create assignment
createAssignment()
save assignment
confirm save
assignment created
Guideline Configuration
upload guidelines
extractGuidelines()
store rules
confirm save
guidelines confirmed
Assignment Submission
submit assignment
checkEligibility()
verify deadline & attempts
eligibility status
format validation
plagiarism check
save submission (Pending)
confirm save
submission acknowledged
show submission status
Evaluation
view submissions
fetch submissions
retrieve submissions
submission data
display submissions
evaluate submission
submit evaluation
save scores & feedback
confirm save
evaluation completed
Result Publication
finalize result
update final score
confirm update
publish result
view final result
Administration
view system reports
fetch analytics
report data
show system statistics

SYSTEM ARCHITECTURE



LIMITATIONS

- Compliance checking is restricted to **predefined rules and templates**
- Semantic quality of content is **not fully automated**
- Plagiarism detection is limited by **available reference databases**
- System performance depends on **document size and complexity**
- Initial guideline setup requires **manual effort by teachers**
- Real-time scalability is limited in **non-cloud deployments**

FUTURE SCOPE

- Integration of **AI-based semantic and content quality analysis**
- Advanced **plagiarism detection** using larger academic repositories
- Support for **multiple document formats and regional languages**
- **Cloud-based deployment** for scalability and availability
- Real-time **analytics and performance dashboards**
- Integration with **Learning Management Systems (LMS)**

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EduDOC

Login

Teacher Register

Intelligent Academic Compliance Platform

EduDOC delivers structured guideline enforcement, plagiarism intelligence, transparent audit reports, and annotated academic feedback — enabling institutions to maintain academic integrity at scale.

Launch Platform

Register as Educator

THANK YOU

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